Frontend Overview

The frontend is a React 19 single-page application that serves as the primary interface for ICT staff and administrators.

Technology Stack

Technology	Version	Purpose
React	19.2	UI framework
Vite	7.2	Build tool and dev server
Tailwind CSS	3.4	Utility-first CSS
Lucide React	0.554	Icon library
Recharts	3.4	Data visualization
React Router DOM	Latest	Client-side routing
React Hot Toast	Latest	Toast notifications

Entry Points

main.jsx

```
import React from 'react';
import ReactDOM from 'react-dom/client';
import App from './App.jsx';
import './index.css';

ReactDOM.createRoot(document.getElementById('root')).render(
  <React.StrictMode>
    <App />
  </React.StrictMode>
);
```

App.jsx

The root component manages:

- Authentication state
- Active tab/navigation
- Global data fetching (tickets, statistics)
- Branding application from public settings

Tab-Based Navigation

The authenticated interface uses a tab-based layout:

Tab	Component	Role Access
Dashboard	`DashboardView`	Staff, Admin
Tickets	`TicketsView`	Staff, Admin
Departments	`DepartmentsView`	Staff, Admin
Users	`UsersView`	Staff, Admin
Knowledge Base	`KnowledgeBaseView`	Staff, Admin
Admin	`AdminPanel`	Admin only

Data Fetching Strategy

The application uses a custom hook-based data fetching pattern:

Component Hierarchy

State Management

The application uses React's built-in state management without external libraries:

- **Local State:** `useState` for UI state (modals, filters, forms)

- **Derived State:** `useMemo` for expensive calculations (filtered ticket lists)
- **Side Effects:** `useEffect` for data fetching and branding application
- **Persistence:** `localStorage` for JWT token and user session

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Troubleshooting

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